Title: The use of color in foraging in Lessert's Rainbow Spider (*Stenaelurillus lesserti)*

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Intro:

Evolutionary processes play a crucial role in driving the sensory adaptation of organisms to their environments, leading to the emergence of complex traits (Dangles and Casas 2019; Endler 1986). The evolution of such traits in different species provides compelling evidence of the power of natural selection to shape sensory systems according to ecological needs (Dawkins 2009). Color vision, specifically, is one such trait that is relatively common across many animals (Gerl & Morris 2008; Briscoe and Chittka 2001; Kelber & Jacobs 2016). Enhanced color vision can help animals detect food, predators, distinguish among mates, and detect and discriminate other vital environmental cues (Cronin et al. 2014). Surprisingly, color vision is also known to have evolved independently multiple times within the same taxonomic group (Yokoyama 2008; Jacobs 2009). These independent origins of color vision within the same taxonomic group provide a case to test the hypotheses on the reasoning behind the evolution of color vision. Identifying such taxonomic groups and studying them can provide us with incredible opportunities to get a better understanding of the mechanism of why color vision has evolved in the way it has.

The evolution of color vision in the family Salticidae, or jumping spiders, is a particularly intriguing example in this context. Salticidae is a large family of spiders including more than 6500 species. Jumping spiders, which are highly visual predators, are known for their exceptional visual acuity (Williams & Melntyre, 1980), and recent work suggests that color vision has evolved independently at least three (and likely more) times within this group (Zurek et al 2015, Outomuro et al. 2019, Cross et al. 2020). Most salticids have only UV and green photoreceptors which is known as dichromatic vision but some salticids have evolved enhanced color vision by either adding filter pigments (as in the genus *Habronattus, Zurek et al 2015*) or by adding additional photoreceptors in the longer wavelength range (as is the case in *Stenaelurillus* and *Maratus* which each have a total of four photoreceptors, making them tetrachromatic, see Outomuro et al. 2019). In the genus *Habronattus*, a red spectral filter has evolved that allows spiders to see and discriminate long wavelength coloration (e.g., reds and oranges) that experiments suggest may be used for assessing colorful mates as well as avoiding aposematic prey (Taylor and McGraw 2013; Taylor et al. 2014; Zurek et al. 2015). In the genus *Evarcha* (with a currently unknown mechanism of color vision), color manipulation experiments have shown that spiders use red coloration to identify their preferred prey of blood-carrying mosquitoes (Cross et al. 2020). However, the roles that color vision plays in other species with independent origins of color vision (e.g., *Stenaelurillus, Maratus*) have not been explored. The diversity and advancement in color vision potentially help these spiders to see and discriminate red from other colors during foraging (Taylor et al. 2016), and may also improve their ability to detect colorful prey against naturally-colored backgrounds, depending on their color vision, compared to other species that do not have color vision. Understanding the benefits of color vision across these different species within Salticidae could be critical in understanding the independent evolution of color vision across different species of jumping spiders. This, in turn, could be crucial to elucidate broader evolutionary patterns like convergent evolution, where similar traits arise separately in different lineages, highlighting the influence of environmental variations in shaping sensory systems (Li & Zhang, 2021). This phenomenon highlights the remarkable adaptability of organisms and provides insights into the mechanisms of natural selection.

While color vision has evolved repeatedly across salticids, we are still unclear on why this is so. Color vision in salticids may have evolved in either foraging or mating contexts, or a combination of both. Vivid coloration can be used to attract mates or intimidate rivals, with visual signals being a key component of mating displays and courtship behaviors (Taylor & McGraw, 2013). Additionally, in the context of foraging, a potential hypothesis that has been studied in different salticid systems is that color vision might provide foraging benefits. For instance, in *Habronattus*, there is evidence to suggest that color vision may aid in foraging in two possible ways. First, since *Habronattus* are generalist predators and feed on a variety of small arthropods, color vision allows them to identify and avoid toxic aposematic prey (Powell, Stalcup & Taylor, In Prep). Second, color vision can improve foraging success by allowing them to identify non-toxic prey in colored backgrounds (Lietzenmayer et al, In prep). Similar to *Habronattus*, color vision in other salticids may have also evolved in a foraging context. However, understanding the primary drivers of color vision evolution requires detailed examination of other specific salticid species and their ecological interactions.

Stenaelurillus lesserti, is a particularly compelling subject for this type of research (Figure 1). This species has been known to have a diet significantly based on termites, but they also occasionally prey upon other insects in both the field and lab (personal observation, see also Hill et al., 2024). This species inhabits environments with predominantly green grass, dry twigs, sand, and termite mounds where color vision could provide substantial foraging advantages to detect prey within these colorful backgrounds. Additionally, they have tetrachromatic vision which means they have 4 different photoreceptors. Hence, by focusing on the foraging context in this species, we can test our hypotheses further about the adaptive benefits of color vision in prey detection and capture as compared to other species of jumping spiders. This approach allows for a detailed analysis of how visual capabilities enhance foraging success, thereby investigating the hypothesis that color vision evolved independently across different jumping spider species in response to ecological pressures related to food acquisition.

Figure 1: Images of male (left) and female (right) *Stennaelurillus lesserti.* Note that the male has bright red and blue coloration on his face that is not present in females. Creative Commons images from iNaturalist users Vipin Baliga (left) and Uma Vaijnath (right) (CC-BY-NC-ND).

The goal of the present study was to determine if the tetrachromatic color vision observed in *S. lesserti* enhances foraging by improving their ability to hunt for prey in natural backgrounds. In this study, we first determined whether the spiders have a natural bias for or against particular colors. We then went on to conduct a foraging experiment to assess whether the ability to use color cues (i.e., chromatic contrast) improved the spiders' foraging efficiency. Our inferences from these experiments could reveal significant aspects of sensory evolution and ecological adaptation in jumping spiders. Furthermore, a comparative analysis of the visual systems of *S. lesserti* with other jumping spider species that possess or lack color vision can infer the evolutionary pathways and selective pressures involved. Such comparisons can elucidate whether similar ecological demands have led to convergent evolution (Morehouse et al., 2017).

Methods

In order to understand if color vision in *S. lesserti* improves foraging success, we designed a study in two phases. In the first phase, our goal was to confirm that there were no natural prey color biases in this species (as there are in other jumping spider species, see Taylor et al. 2014, Powell et al. 2019). In the second phase, our goal was to understand if the ability to use color cues confers a foraging benefit when the spiders are searching for colorful prey in natural environments.

These methods were modified from a related study that assessed the fitness benefits of color vision in a different species of jumping spider, *Habronattus pyrrithrix* (Lietzenmayer et al, In prep). To maximize the objectivity of these studies, we pre-registered our hypothesis, protocols, sample sizes, and planned data analyses (hereafter referred to as 'confirmatory analyses') with the Center for Open Science before the start of any data collection on *H. pyrrithrix* (Lietzenmayer et al. 2023, see Winsor et al. 2020 for the rationale behind preregistering studies before data collection). The current study on *S. lesserti* builds on this previously preregistered study to test the same *a priori* hypotheses and directional predictions. We modified some methods for logistical reasons (described below).

Spider Collection and Maintenance

We collected *S. lesserti* on the campus of Azim Premji University, Bengaluru, India. Spiders were fed termites (*Odontotermes obesus*) of approximately 75% of the spiders' body size every other day. Spider housing enclosures were made with clear plastic cups (8 cm diameter) with a small hole (2 cm diameter) at the top for ventilation. This hole was plugged with cotton and sprayed with water daily to provide moisture. We used a mix of cut pieces of brown paper and natural leaves/grass inside the enclosure for substrate and enrichment (Carducci and Jakob 2000). For both phases of the study, we grouped the spiders into males, females, and juveniles to see if there is any behavioral variation based on either sex or stage of spiders. We grouped the juvenile spiders together irrespective of the sex as they do not show any sexual dimorphism prior to attaining maturity, which makes them unreliable to be sexed (Bartos, 2008; Lim & Li, 2007; Nelson, Jackson, & Sune, 2005).

Prey Color Manipulation

We hand-collected termites (*Odontotermes obesus*) from natural termite mounds on the Azim Premji University Campus, and we used the termites in experiments on the same day that they were collected. We provided them with moist filter paper and rotten wood as substrate during the time between collection and the start of experiments.

After attacking termites, *S. lesserti* typically feeds on the abdomen of the termite first (pers. obs.). Hence, for both stages of experimentation, we painted the abdomen of the termites with the colors required for each experiment. For all base colors and color mixtures, we used gloss enamel paint (Testors, Rust-Oleum, Vernon Hills, IL, USA). These termite-painting methods have been used successfully in several previous experiments with jumping spiders and the paints have no detectable effect on termite behavior (Vickers et al, 2021; Vickers and Taylor, 2020; Windsor et al, 2020; Powell et al, 2019; Vickers and Taylor, 2018).

For the first stage of experiments (where we were assessing initial color biases), we used four base colors of paint: blue, red, yellow and green. For the second stage of experiments (where we were assessing whether the ability to use color cues provides a foraging benefit), we prepared color mixtures of red, orange, and green of different brightness levels. In this stage, the green mixture consisted of 1:2 ratio of green to yellow paint and the orange mixture consisted of 1:3 ratio of red to tangerine paint (names provided on Testors paint jars). The red color mixture used here was unmanipulated from the original red. The brightness levels were manipulated by adding gloss white or gloss black to either lighten or darken the shades. We used a UV-vis spectrophotometer (USB 2000 with PX-2 pulsed xenon light source, Ocean Optics, Dunedin, FL) to measure 10 samples of each color at each brightness level for ensuring that each set of colors were statistically similar within each brightness shade (light green=light orange=light red) and different among brightness levels (for example, light green > medium green > dark green). Overall brightness calculations were taken within the spiders' visible range of 280-700 nm (Peaslee and Wilson 1989; Li et al. 2008). Upon painting the termites, we waited a minimum of 10 minutes prior to presenting the spiders with termites to ensure that the paint was fully dry and any volatiles from the paint had dissipated.

Laboratory Conditions

We conducted all experiments in a room with ample sunlight (through many large windows) in addition to full spectrum lights (City Greens Aluminium 24W Full Spectrum Electric Led Grow Lights, City Greens, Bangalore, India) mounted directly over the experimental area. The temperature and humidity were not altered and hence, were the same as the ambient conditions that the spiders are acclimated to. Additionally, all the experiments were conducted and completed between 8 am to 5 pm, which are known to be the active hours for these spiders (Pratush Brahma, personal observations).

Phase 1: Identifying Natural Color Biases

Some jumping spider species (e.g., *Habronattus pyrrithrix*) have natural aversions to certain prey colors (Taylor et al. 2014). To identify any such natural biases in *S. lesserti*, we collected a total of 30 individuals (11 adult males, 9 adult females and 10 juveniles) between November 29 and December 25, 2024. These spiders were housed in plastic cups as mentioned above in the laboratory for 2 days prior to testing so that they could acclimate to lab conditions. Additionally, they were unfed until the day of testing to ensure the spiders were hungry enough to eat a termite and hence, potentially make a choice. To ensure consistency of hunger levels, we would only collect spiders between 8 am -12 pm on the collection day. Assuming the collection day to be Day 1, we would test the spiders collected on Day 1 on Day 3. This was repeated for all the individuals involved in this test.

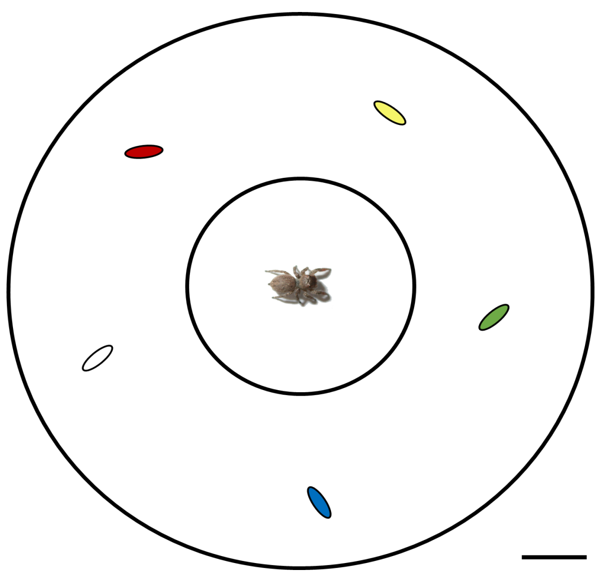


Figure 2: Experimental Setup for phase 1: *Figure indicative of an experimental setup with the spider enclosure (small petri dish) at the center from where the focal spider is released to the larger petri dish with 4 termites that are painted on their abdomen with the colors blue, red, green, yellow and an unpainted termite. Figure The scale bar represents 1 cm.*

We created five color groups of termites using the methods described above: red, yellow, green, blue, and unpainted (with a natural color of light yellow/white). We chose these color groups because they span the color range of possible insect prey for *S. lesserti* that they may encounter in the field (Pratush Brahma, Dr. Divya Uma, personal observations). For prey color bias tests, each spider was kept in an enclosed small petri dish (diameter: 35 mm, height: 10 mm). This enclosure was then placed at the center of a larger petri dish (diameter: 94 mm, height: 16 mm) (**Figure 2**). The larger petri dish had a slightly moist filter paper at its base to provide a substrate for the termites to grip and move around. Upon placing the smaller petri dish with the spider in the larger petri dish, we put the 4 colored and 1 uncolored termites in the larger petri dish surrounding the spider enclosure. The spider was kept enclosed for the next 2 minutes prior to releasing them in the arena to get them acclimated to the setup. During the acclimation period, the spider could easily observe the termites through the transparent walls of the petri dish. The entire experimental arena was surrounded by white paper to provide a uniform background and to prevent visual distractions. Upon completion of the acclimation time, we removed the lid of the smaller petri dish to release the spider into the larger petri dish where it could freely interact with the termites. We then quickly covered the lid of the larger petri dish to ensure that the spider did not escape the experimental arena. Within the larger petri dish, there was enough space between the top of the smaller petri dish and the top of the lid of the larger petri dish, for the spider to be able to crawl through once the larger lid is placed. The spiders were then allowed to forage on the termites for 30 min.. For each spider, we recorded all the termites attacked during the 30-min trial. All trials were conducted under the full spectrum light as mentioned above and during the known active periods for the spiders (8 am - 5 pm IST). We marked a successful trial only upon observing at least a single attack from each spider in the recording. If a spider did not attack any termite, we repeated the test again on Day 4 of collection. Retested spiders remained unfed to maintain a hunger level conducive to making an active choice in the trial. We released the spiders back into the wild upon either completion and verification of a successful trial in the first attempt or at the end of the second attempt, irrelevant of successful or unsuccessful trial. We recorded which color of termite the spider first oriented to and which color they first attacked. Spider orientation towards termites is the observed behavior where a spider orients its body to directly face a specific termite, allowing them to inspect it with their large forward-facing eyes, a response clearly identifiable within our experimental observations. We only included successful trials (i.e., trials in which the spider attacked at least one termite) for statistical analysis.

Phase I Statistical Analysis:

We used likelihood ratio X^2 tests to assess whether the spiders showed any biases in (1) the termite color that they first oriented to (i.e., the termite that first got their attention) or (2) the termite color that the spider attacked. All data were analyzed using R.

Phase 2: Assessing whether Color Vision Improves Foraging

To understand if color vision in *S. lesserti* improves their foraging, we collected a total of 60 individuals (20 adult males, 20 adult females and 20 juveniles) between January 10 and Feb 12, 2024. These spiders were housed in plastic cups as mentioned above in the laboratory for 3 days prior to testing for acclimation to lab conditions. Additionally, they were unfed until the day of testing to ensure the spiders were hungry enough to actively forage. To ensure consistency of hunger levels, we would only collect spiders between 8am -12 pm of the collection day. Assuming the collection day to be Day 1, we would test the spiders collected on Day 1 on Day 4. This was repeated for all the individuals involved in this test.

We randomly assigned each spider to one of two treatment groups with different foraging scenarios (i.e., single-color vs. multicolor). Both the treatment groups consisted of a clear plastic arena (10 cm X 10 cm X 5 cm) with a mesh hole for ventilation on the side (2 cm diameter) and a light brown fabric substrate covered with 10 grams of evenly spaced green plastic fake foliage (Ashland Fern Collection, Michael's Stores, Irving, TX, USA). Each arena had a circular hole (1 cm diameter) directly in the middle of the top panel for introducing termites and the spider.

In the single-color treatment, we gave the spiders a foraging environment that only consisted of green-colored termites on the background of green foliage (absence of chromatic cues to allow the spider to distinguish the termites from the foliage). Here, the spiders were presented with nine green termites, three each for different levels of brightness (light green, medium green, dark green; manipulation described above). In the multicolor treatment, spiders had the same green background as the single-color treatment, but the termite prey chromatically contrasted with the background. We gave the spiders in the multicolor treatment nine termites as well, but instead of all green, we included three green, three orange, and three red termites (one each for different levels of brightness). This allowed for both treatment groups to have three termites at each brightness level, regardless of color. In both treatments, spiders had two hours to forage on these termites before the trial ended. Each spider was tested only once in this trial prior to releasing them back in the wild. We video recorded the entire foraging trial for each spider. The behaviors we recorded to assess whether color vision improves foraging success when color cues are present included (1) the amount of time it took for spiders to make their first attack (in seconds), (2) the total number of termites attacked and consumed during the 2-hour trial. We also recorded the color and brightness of the termite first attacked and the color and brightness of all termites attacked within the two-hour trial. All trials took place within natural daylight hours between 8am and 5pm IST in front of large windows supplemented with overhead full-spectrum light bulbs (City Greens Aluminium 24W Full Spectrum Electric Led Grow Lights, City Greens, Bangalore, India).

Phase 2 Statistical Analysis:

To assess whether spiders had improved foraging success when they had access to color cues, we used 1-tailed t-tests (or nonparametric Mann-Whitney U tests) to test the directional predictions that spiders in the multicolor treatment group (1) were quicker to make their first attack and (2) attacked and ate more termites overall compared to spiders in the single-color treatment group. We used one-tailed tests because our hypotheses led to explicitly directional predictions, and these analyses were planned and pre-registered with the Center for Open Science prior to any data collection (see above).

Further exploratory analyses:

Because we found that the spiders had no initial color biases when they were foraging for termites in an open petri dish with a white background (see Results of Phase 1), any biases in attack color by the multicolor treatment group in Phase 2 of the experiment is likely to be driven by how well the spiders could see the different termite colors among the green foliage. If color vision improves the spiders' ability to find colored prey on colored backgrounds, we would expect the spiders in the multicolor treatment group to attack red and orange termites at higher rates compared with green termites. We used two paired t-tests to test this prediction, comparing each spider's attack rate on green vs. red and green vs. orange.

We also assessed whether the spiders had any preference for brightness of the prey (irrespective of the color). We used ANOVA to ask whether the brightness categories of the colors (light, medium, dark; regardless of the color) predicted the number of termites attacked. Because each spider had a value for light, medium, and dark, spider ID was included as a random effect in the model.

Results

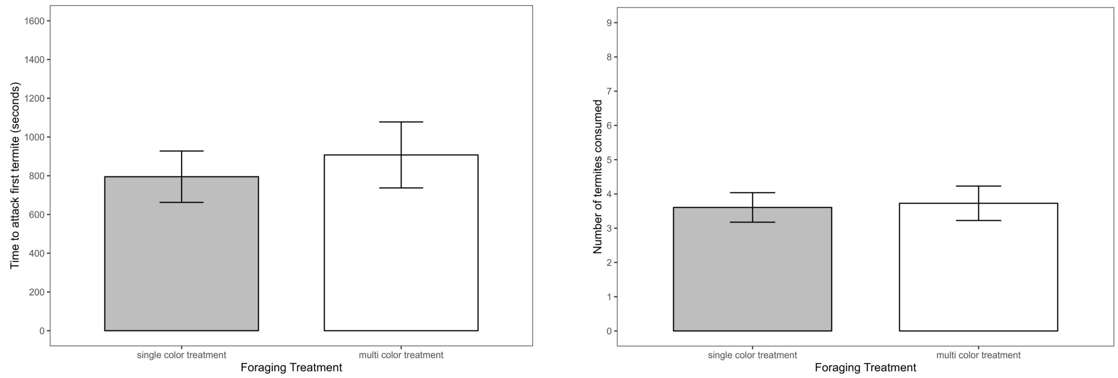
Phase 1

When foraging for colored termites in an empty petri dish with a white filter paper background, the spiders did not show any biases in either the color that first got their attention (i.e. the color that they first oriented to) (likelihood ratio X^2=4.10, P=0.39) or the color that they first attacked (likelihood ratio X^2=1.36, P=0.851).

Phase 2

Contrary to our predictions, spiders in the multicolor treatment group were not quicker to make their first attack (t=0.521, P=0.698, Figure 3a) and did not attack and eat more termites overall (t=0.183, P=0.428, Figure 3b) compared to spiders in the single-color treatment group.

For our exploratory analysis, within the multicolor treatment, the spiders were not more likely to attack red or orange termites compared with green (Figure 4, red vs. green: t=-1.468, P=0.152; orange vs. green (t=0.312, P=0.757). Moreover, the spiders did not show any preference for any of the brightness levels (light, medium, dark) compared with others ($F_{2,130}$ = 0.1957, P = 0.8225).



(a) (b)

Figure 3: Experimental results assessing whether spiders in the multicolor treatment group ("color cues present") have higher foraging success compared with spiders in the single-color treatment group ("color cues absent"). Measures of foraging success were (a) the time it took the spider to attack the first termite, and (b) the total number of termites consumed during the 2-hour trial. Bars represent means ± SE.

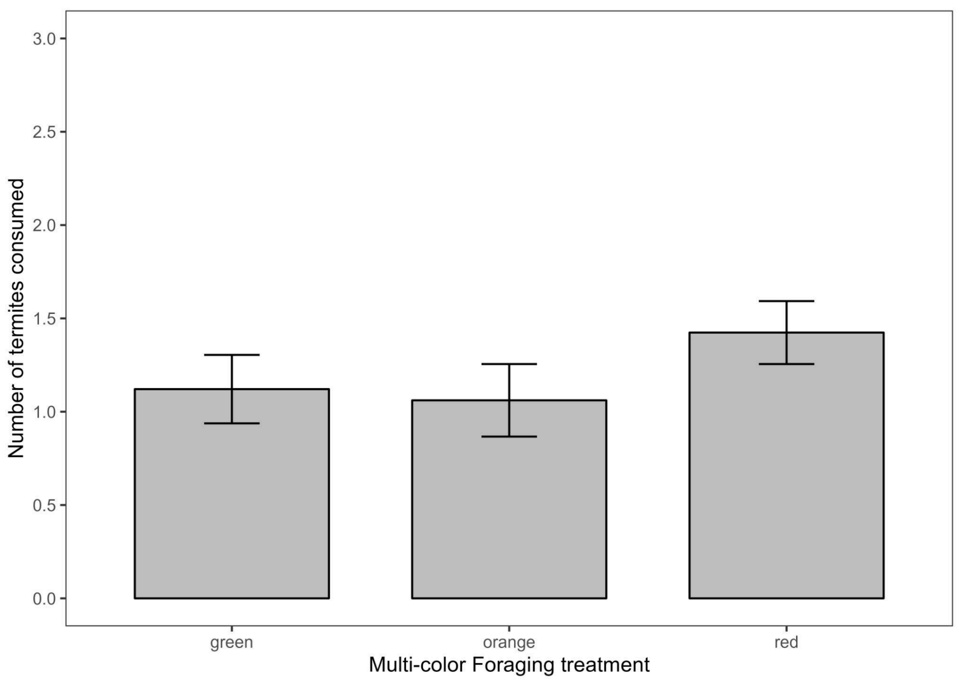


Figure 4: Results of exploratory analyses that compare the mean numbers of colored termites consumed in the multi-colored treatment. Contrary to expectation, the green termites were not consumed at lower rates compared with either red or orange. Bars represent means ± SE.

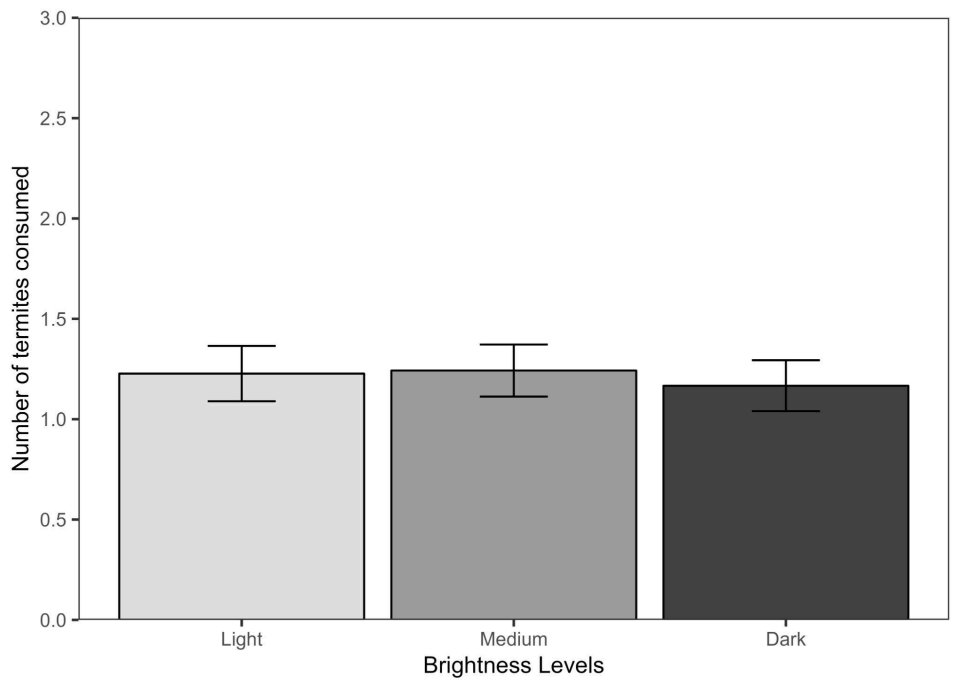


Figure 5: Results of exploratory analyses that compare the mean numbers of colored termites consumed across the different brightness levels (light, medium and dark). There was no significant difference between the number of termites consumed between the brightness levels. Bars represent means ± SE.

Discussion

Our study investigated the role of tetrachromatic color vision in the foraging behavior of a termite-specialist jumping spider, *Stenaelurillus lesserti*. We wanted to specifically understand if color vision enhances prey detection against natural backgrounds. Our results suggest that enhanced color vision of *S. lesserti* does not confer a significant advantage in detecting colorful prey on a naturally colored background, at least in the context that we tested them. This contrasts with findings in other jumping spiders, such as *Habronattus pyrrithrix* where color vision evidently aids in foraging decisions through improved prey detection (Lietzenmayer et al, In prep). Also, in the case of *Evarcha culicivora*, where color vision is shown to aid foraging decisions by either avoidance of aposematic prey or identification of preferred prey (Cross et al., 2020; Lietzenmayer et al., in prep; Powell et al., in prep). It may be that the unique dietary preference of *S. lesserti* on termites may explain why they seem to rely less on color cues compared with other species.

The lack of a foraging benefit from color cues was evident from our Phase 2 trials. Spiders that were given the opportunity to rely on color cues to detect and capture prey (i.e., in the multicolor treatment) did not exhibit improved foraging efficiency compared to spiders that could only rely on achromatic cues (i.e., in the single-color treatment). Specifically, there was no significant difference in the time taken to make the first attack (Figure 3a) or in the total number of termites attacked and consumed between the two treatments (Figure 3b). Additionally, exploratory analysis within the multicolor treatment showed no significant preference for attacking red or orange termites over green termites, even though the green termites should blend in with their green background (Figure 4). This collective evidence suggests that the ability to use color cues when foraging, as manipulated in our study, was not a critical factor driving foraging success for *S. lesserti* in this context.

Our conclusion that color vision may be less critical for *S. lesserti's* foraging decisions compared to other species was also strengthened through our Phase 1 results. The spiders in the Phase 1 study showed no statistically significant bias in their initial orientation towards, or their first attack on the termites painted with various colors (red, yellow, green, blue vs. unpainted). This absence of prey color bias contrasts sharply with the documented aversion to red/yellow in the generalist predator *Habronattus*, likely linked to avoiding toxic prey (Taylor et al., 2014), and the preference for red in the mosquito-specialist *Evarcha culicivora*, linked to identifying red blood-filled mosquitoes (Cross et al., 2020). The lack of such biases in *S. lesserti* too may make sense given its diet preference on termites, which are typically drab and cryptic in coloration (often varying in shades of white to light brown). In this context, it may not make sense to have biases for some colors over others. While they likely encounter (and even prey on) colorful prey occasionally, these interactions are likely less frequent and less important in shaping any foraging biases compared with their interactions with termites (personal observation, see also Hill et al., 2024).

There are several possibilities that might explain why *S. lesserti*'s tetrachromatic vision might not enhance foraging success in the way observed in other salticids. As a termite specialist, *S. lesserti* may prioritize alternative sensory cues over color. For example, they might rely more heavily on achromatic visual cues like movement, shape, texture, or brightness contrast, which could be more relevant for detecting camouflaged or subtly colored termites. Cues associated with termite mounds or trails, not included in our study, might also play a significant role in natural foraging. These spiders may also rely on non-visual cues (e.g., vibrations, olfactory cues, etc) when foraging for termites. The advanced color vision of *S. lesserti* may primarily serve functions outside of foraging. The pronounced sexual dimorphism, with males displaying vibrant red and blue facial coloration absent in females (Figure 1), potentially implicates color vision in intraspecific interactions, such as mate choice or rival assessment (e.g., Lim & Li, 2004; Taylor & McGraw, 2013). Navigation or predator avoidance are other potential contexts where tetrachromacy could be advantageous, and should be examined further.

Despite its multiple independent evolutionary origins (Zurek et al., 2015; Outomuro et al., 2019; Cross et al., 2020), enhanced color vision in jumping spiders may serve diverse, context-dependent functions rather than having a uniform function across species. The ecological niche, particularly foraging strategy (specialist vs. generalist) and prey characteristics, appears crucial in shaping how color vision is used. Similarly, responses to achromatic patterns on prey-like stripes also vary across species, influencing attention and attack decisions (Gawel et al 2023, Humbel et al., 2023). This study underscores that possessing an enhanced sensory capability, like tetrachromacy, does not guarantee its use in all potential ecological scenarios, such as discriminating prey based on color contrast against a background.

Future research should explore the potential roles of color vision in other areas of *S. lesserti* ecology, particularly mating behavior, given the striking sexual dimorphism (Figure 1). Investigating foraging behavior in more natural settings, perhaps by incorporating termite mounds or trails, and explicitly testing the relative importance of other visual cues (e.g., motion, shape) versus color could yield deeper insights into the sensory ecology of this predator. Ultimately, comparative studies encompassing a wider range of salticid species with varied diets, visual systems, and other aspects of natural history (Humbel et al., 2023) will be vital for fully understanding the complex evolutionary trajectories and diverse functions of color vision within this remarkable group of spiders.

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